

An Analysis of Mathematical optimization

Mathematical optimization

Mathematical optimization -- Part 1

Sensitivity and continuity of optima The envelope theorem describes how the value of an optimal solution changes when an underlying parameter changes. The process of computing this change is called comparative statics. The maximum theorem of Claude Berge (1963) describes the continuity of an optimal solution as a function of underlying parameters.

Mathematical optimization -- Part 2

"Decision Tree for Optimization Software". Links to optimization source codes "Global optimization". Archived from the original on 2022-01-29. Retrieved 2019-05-18. "EE364a: Convex Optimization I". Course from Stanford University. Varoquaux, Gaël. "Mathematical Optimization: Finding Minima of Functions".

Mathematical optimization -- Part 3

$f(x_0) \geq f(x) \Leftrightarrow -f(x_0) \leq -f(x)$, $\{\displaystyle f(\mathbf{x}_0) \geq f(\mathbf{x}) \} \Leftrightarrow -f(\mathbf{x}_0) \leq -f(\mathbf{x})$

Mathematical optimization -- Part 4

asks for the maximum value of the objective function $2x$, where x may be any real number. In this case, there is no such maximum as the objective function is unbounded, so the answer is "infinity" or "undefined".

Mathematical optimization -- Part 5

Given: a function $f : A \rightarrow \mathbb{R}$ from some set A to the real numbers Sought: an element $x_0 \in A$ such that $f(x_0) \leq f(x)$ for all $x \in A$ ("minimization") or such that $f(x_0) \geq f(x)$ for all $x \in A$ ("maximization"). Such a formulation is called an optimization problem or a mathematical programming problem (a term not directly related to computer programming, but still in use for example in linear programming – see History below). Many real-world and theoretical problems may be modeled in this general framework. Since the following is valid:

Mathematical optimization -- Part 6

An optimization problem with discrete variables is known as a discrete optimization, in which an object such as an integer, permutation or graph must be found from a countable set. A problem with continuous variables is known as a continuous optimization, in which optimal arguments from a continuous set must be found. They can include constrained problems and

multimodal problems. An optimization problem can be represented in the following way:

Mathematical optimization -- Part 7

Economics and finance Economics is closely enough linked to optimization of agents that an influential definition relatedly describes economics qua science as the "study of human behavior as a relationship between ends and scarce means" with alternative uses. Modern optimization theory includes traditional optimization theory but also overlaps with game theory and the study of economic equilibria. The Journal of Economic Literature codes classify mathematical programming, optimization techniques, and related topics under JEL:C61-C63. In microeconomics, the utility maximization problem and its dual problem, the expenditure minimization problem, are economic optimization problems. Insofar as they behave consistently, consumers are assumed to maximize their utility, while firms are usually assumed to maximize their profit. Also, agents are often modeled as being risk-averse, thereby preferring to avoid risk. Asset prices are also modeled using optimization theory, though the underlying mathematics relies on optimizing stochastic processes rather than on static optimization. International trade theory also uses optimization to explain trade patterns between nations. The optimization of portfolios is an example of multi-objective optimization in economics. Since the 1970s, economists have modeled dynamic decisions over time using control theory. For example, dynamic search models are used to study labor-market behavior. A crucial distinction is between deterministic and stochastic models. Macroeconomists build dynamic stochastic general equilibrium (DSGE) models that describe the dynamics of the whole economy as the result of the interdependent optimizing decisions of workers, consumers, investors, and governments.

Mathematical optimization -- Part 8

This represents the value (or values) of the argument x in the interval $(-\infty, -1]$ that minimizes (or minimize) the objective function $x^2 + 1$ (the actual minimum value of that function is not what the problem asks for). In this case, the answer is $x = -1$, since $x = 0$ is infeasible, that is, it does not belong to the feasible set. Similarly,

Mathematical optimization -- Part 9

Necessary conditions for optimality One of Fermat's theorems states that optima of unconstrained problems are found at stationary points, where the first derivative or the gradient of the objective function is zero (see first derivative test). More generally, they may be found at critical points, where the first

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derivative or gradient of the objective function is zero or is undefined, or on the boundary of the choice set. An equation (or set of equations) stating that the first derivative(s) equal(s) zero at an interior optimum is called a 'first-order condition' or a set of first-order conditions. Optima of equality-constrained problems can be found by the Lagrange multiplier method. The optima of problems with equality and/or inequality constraints can be found using the 'Karush–Kuhn–Tucker conditions'.

Mathematical optimization -- Part 10

Operations research Another field that uses optimization techniques extensively is operations research. Operations research also uses stochastic modeling and simulation to support improved decision-making. Increasingly, operations research uses stochastic programming to model dynamic decisions that adapt to events; such problems can be solved with large-scale optimization and stochastic optimization methods.

Mathematical optimization -- Part 11

Mechanics Problems in rigid body dynamics (in particular articulated rigid body dynamics) often require mathematical programming techniques, since you can view rigid body dynamics as attempting to solve an ordinary differential equation on a constraint manifold; the constraints are various nonlinear geometric constraints such as "these two points must always coincide", "this surface must not penetrate any other", or "this point must always lie somewhere on this curve". Also, the problem of computing contact forces can be done by solving a linear complementarity problem, which can also be viewed as a QP (quadratic programming) problem. Many design problems can also be expressed as optimization programs. This application is called design optimization. One subset is the engineering optimization, and another recent and growing subset of this field is multidisciplinary design optimization, which, while useful in many problems, has in particular been applied to aerospace engineering problems. This approach may be applied in cosmology and astrophysics.

Mathematical optimization -- Part 12

Adding more than one objective to an optimization problem adds complexity. For example, to optimize a structural design, one would desire a design that is both light and rigid. When two objectives conflict, a trade-off must be created. There may be one lightest design, one stiffest design, and an infinite number of designs that are some compromise of weight and rigidity. The set of trade-off designs that improve upon one criterion at the expense of another is known as the Pareto set. The curve created plotting weight against stiffness of the best designs is known as

the Pareto frontier. A design is judged to be "Pareto optimal" (equivalently, "Pareto efficient" or in the Pareto set) if it is not dominated by any other design: If it is worse than another design in some respects and no better in any respect, then it is dominated and is not Pareto optimal. The choice among "Pareto optimal" solutions to determine the "favorite solution" is delegated to the decision maker. In other words, defining the problem as multi-objective optimization signals that some information is missing: desirable objectives are given but combinations of them are not rated relative to each other. In some cases, the missing information can be derived by interactive sessions with the decision maker. Multi-objective optimization problems have been generalized further into vector optimization problems where the (partial) ordering is no longer given by the Pareto ordering.